Practical 8:

Aim : A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

```
  GNU nano 8.7                                   lru.c
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frame[4], time[4];
    int i,j,n=12,frames=4,hit=0,fault=0,counter=0,pos,min;

    for(i=0;i<frames;i++)
        frame[i] = -1;

    for(i=0;i<n;i++) {
        int found = 0;

        for(j=0;j<frames;j++) {
            if(frame[j] == pages[i]) {
                hit++;
                counter++;
                time[j] = counter;
                found = 1;
                break;
            }
        }

        if(!found) {
            fault++;

            if(i < frames) {
                frame[i] = pages[i];
                counter++;
                time[i] = counter;
            }
            else {
                min = time[0];
                pos = 0;

^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute     ^C Location    M-U Undo
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo
```

```
GNU nano 8.7                                    lru.c
            }
        }

        if(!found) {
            fault++;

            if(i < frames) {
                frame[i] = pages[i];
                counter++;
                time[i] = counter;
            }
            else {
                min = time[0];
                pos = 0;

                for(j=1;j<frames;j++) {
                    if(time[j] < min) {
                        min = time[j];
                        pos = j;
                    }
                }

                frame[pos] = pages[i];
                counter++;
                time[pos] = counter;
            }
        }
    }

    printf("Hits = %d\nFaults = %d\n", hit, fault);
    printf("Hit %% = %.2f\nFault %% = %.2f\n",(float)hit/n*100,(float)fault/n*100);

    return 0;
}

^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute     ^C Location    M-U Undo
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo
```

```
F0IN@LAPTOP-MC73RDEH MSYS ~
$ nano lru.c

F0IN@LAPTOP-MC73RDEH MSYS ~
$ gcc lru.c -o lru

F0IN@LAPTOP-MC73RDEH MSYS ~
$ ./lru
Hits = 4
Faults = 8
Hit % = 33.33
Fault % = 66.67

F0IN@LAPTOP-MC73RDEH MSYS ~
$
```